MID-TERM EXAMINATION

(Course Name: B. Tech.) (Semester: Ist)

(October, 2025) OFF LINE mode

Subject Code: BAI 104	Subject: Programming Fundamentals
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a)Describe various types of error messages in python with example.		1
	b) Explain the purpose of loop structure in a programming language with example.		1
	c) Distinguish between mutable and immutable data types in Python with an example.		3
	d) Explain the concept of slicing in Python and on which data structures it can be applied, give example.		2
	UNIT I		CO Mapping
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) State which of the following python statements are valid and invalid, give reason. i. x = input("enter a number") ii. x = input() iii. x = input("") iv. x = input(" ' ") v. x = input(2) vi. x=input("2" + "3")		3
	b) Describe the working of if-elif-else statements with a flow diagram and a PYTHON code example		1
	c) Explain the concept of object-oriented programming (OOP) and how it applies to Python. Discuss classes, objects, and methods with suitable examples.		2
	UNIT II		CO Mapping
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Differentiate between List, Tuple & dictionary with suitable examples of each.		3
	b) Write a Python program to create a dictionary with three key-value pairs: "onion" with a value of 3, "tomato" with a value of 2, and "potato" with a value of 5. Display the dictionary.		2
	c) Discuss the difference between The remove() and pop() methods for lists in Python, providing examples for each.		2